

AI Infrastructure Cost & Optimization Model

*Quantitative Analysis of Cost, Energy, and Constraint
Tradeoffs in AI Related Systems*

Problem:

With the rapid growth of data centers, they come with large electricity demand, costs, carbon output, and infrastructure constraints. Decision makers must evaluate how pricing, policy and strategy can be used to impact efficiency and cost under uncertainty.

Model Framework

Emissions modeled using:

$$E = Y \times C$$

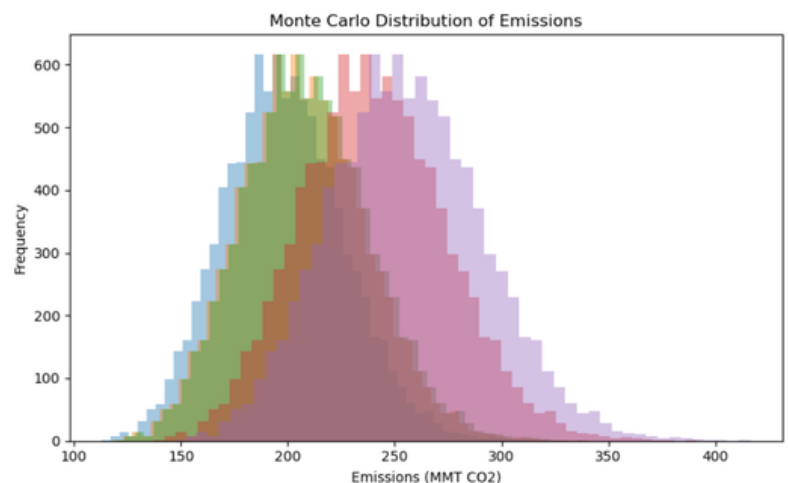
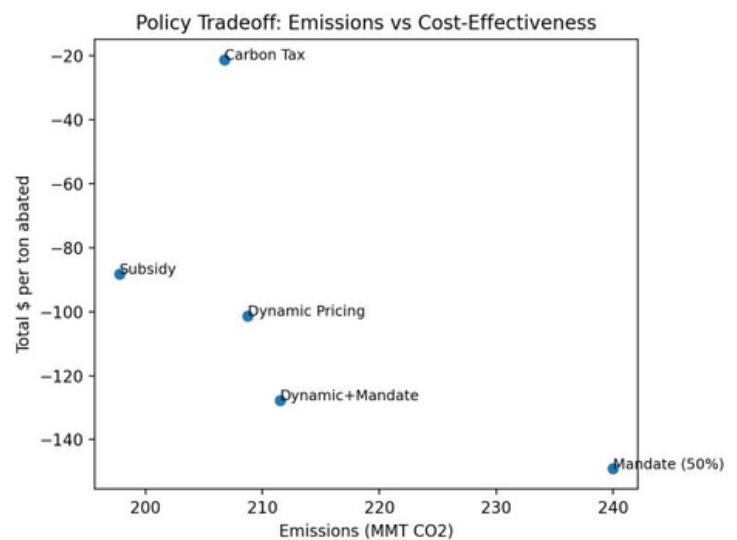
Where electricity demand (Y) and carbon intensity (C) are evaluated across multiple scenarios. The model incorporates multiple projections, +/-10% sensitivity analysis, and a Monte Carlo simulation (10,000 runs) to convey uncertainty in demand growth, pricing and behavior.

Key Findings

- Combination of dynamic pricing and mandate out performed single policy options under uncertainty
- Cost efficient strategies introduced higher volatility

Takeaway

Optimal outcomes are driven by using both price mechanisms and infrastructure constraints in a portfolio strategy that balances cost, efficiency, and risk.



Util – Infrastructure Optimization Platform

A platform for optimising compute deployment across cost, carbon emissions and infrastructure constraints.

- Incorporates live energy pricing and emissions based on location constraints, compute demand, and deployment deadline decisions
- Evaluates the tradeoffs between cost, performance and carbon output
- Designed for real world use in reducing infrastructure monetary and environmental cost
- Involves leading a team of four students in development, research, and market planning



ASSET BRIEFING

Apple Inc. AAPL

Historical market performance, downside risk, and forward scenario analysis prepared from the local finance analytics workbench.

GENERATED 2026-03-20 05:51	COVERAGE 2025-03-19 to 2026-03-19	SOURCE yfinance	RECORDS 252
LATEST PRICE 249.06 USD	OBSERVATION COUNT 252 stored daily records	TOTAL RETURN 16.22% holding-period outcome	
ANNUALIZED RETURN 16.29% geometric equivalent	ANNUALIZED VOLATILITY 31.76% 252-day basis	MAX DRAWDOWN -22.99% peak-to-trough decline	
EXECUTIVE SUMMARY Apple Inc. is presented as an equity briefing based on the locally stored historical price record. Over the covered period, the asset delivered a total return of 16.22% and an annualized return of 16.29%, while realized annualized volatility measured 31.76%. Historical downside metrics indicate a maximum drawdown of -22.99% and a one-day 95% historical VaR of 3.09%.			

Asset Intelligence Workbench – SQL-Based Financial Analysis Platform

A data driven system that integrates market information and external signals for creating simulations for decision support.

- Built a relational database to structure financial and sentiment data, using machine learning to find optimal weighting for each prediction factor
- Outputs a detailed multipaged report that summarises findings, including Monte Carlo simulations, risk metrics and market sentiment reports
- Uses a structured pipeline of data outputs for use in decision making

Financial Risk Engine – Portfolio Simulation Platform

Python-based platform for modeling portfolio risk, return distributions, and downside exposure.

- Implements Monte Carlo simulation and stress testing across portfolios
- Evaluates expected return, volatility, and tail risk
- Designed to support risk-aware capital allocation decisions by involving market simulations for testing

